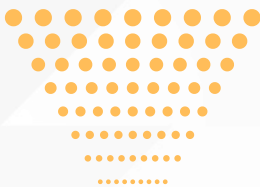




PHILOMATH NETWORKS

**PHILOMATH
NETWORKS PVT. LTD**



6-Days Hands-On Workshop on “Programming Certification Training Course”

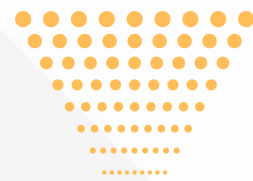
**Get ready to earn your
Programming Certification with us!**

Redefining Learning
Beyond Boundaries



**Phone Number
+91 88698 32559**

About Philomath Networks



At Philomath Networks Pvt. Ltd., we specialize in organizing transformative events for schools, organizations, and communities, fostering a culture of exploration, innovation, and growth. From expert-led workshops to competitive events, we create immersive learning experiences that inspire and empower individuals to go beyond boundaries. Partner with us to bring impactful learning to your community!



Partner With Us

Collaborate with Philomath Networks to bring innovation, engagement, and excellence to your institution or organization. Let's work together to create an ecosystem of lifelong learning and impactful experiences!

What Sets Us Apart?

Comprehensive Learning Approach – We blend academic, non-academic, and skill-based experiences through workshops, competitions, expert-led sessions, and immersive programs tailored to real-world applications.

End-to-End Event Management – From conceptualization to execution, we ensure seamless organization of competitions, tech fairs, hackathons, leadership programs, and customized training sessions that align with institutional goals.

Industry & Academic Bridging – We foster collaborations between institutions, corporates, and experts, ensuring that participants gain practical exposure, networking opportunities, and insights into emerging trends beyond traditional education.

Customizable & Scalable Programs – Whether it's a school-level science fest, university hackathon, corporate leadership training, or specialized certification courses, we tailor our events to meet the unique needs of every institution.

Proven Impact & Engagement – Our interactive approach ensures higher participation, deeper learning retention, and enhanced skill development, making every event a transformative experience.

Course Overview



▶ Course Objective

The Programming Certification Training Course provides a strong foundation in programming, covering core concepts, algorithms, data structures, and software development best practices. Learners will develop skills in writing efficient code, working with databases, and applying object-oriented principles. Through hands-on exercises and project development, they will gain practical experience in building scalable applications. By the end of the course, participants will be able to write optimized, maintainable code and confidently apply for programming roles.

▶ Course Outcome

✓ Master programming fundamentals.

✓ Work with databases and files.

✓ Implement efficient algorithms.

✓ Debug, test, and optimize code.

✓ Apply object-oriented principles.

✓ Build a real-world project.

Training

Modules

The training modules are designed to build a strong programming foundation through a structured learning approach. Starting with fundamental concepts, learners will explore programming principles, control structures, and problem-solving techniques. They will gain hands-on experience with data structures, algorithms, and object-oriented programming, essential for writing efficient and scalable code. Additionally, the course covers debugging, testing, and optimization techniques to enhance coding proficiency.

Module 1: Introduction to Programming

- Understanding the fundamentals of programming and computation.
- Exploring various programming paradigms (Procedural, Object-Oriented, Functional).
- Learning the importance of algorithms and data structures.
- Comparing different programming languages and their applications.
- Understanding software development life cycle (SDLC).

Module 2: Programming Basics

- Learning Python, Java, and C++ syntax and differences.
- Understanding variables, constants, data types, and type conversion.
- Exploring operators (arithmetic, logical, relational, bitwise).
- Writing control structures: loops (for, while) and conditionals (if-else, switch).
- Working with functions: defining, calling, and parameter passing.
- Hands-on: Implementing a basic calculator in Python, Java, and C++.

Module 3: Object-Oriented Programming (OOP)

- Understanding the principles of OOP: Encapsulation, Inheritance, Polymorphism, Abstraction.
- Creating classes and objects in Python, Java, and C++.
- Implementing constructors and destructors for object lifecycle management.
- Exploring method overloading and overriding concepts.
- Understanding static vs. instance members in OOP.
- Hands-on: Developing a simple banking system using OOP concepts.



Training Modules

As learners progress, they will dive into database management, file handling, and real-world application development. The course emphasizes practical implementation through hands-on exercises and project-based learning. By the final module, participants will integrate their skills into a complete project, applying best practices in software development. This structured approach ensures that learners are well-equipped to develop optimized, maintainable, and industry-ready code.

Module 4: Data Structures & Algorithms

- Implementing basic data structures: arrays, linked lists, stacks, and queues.
- Understanding searching algorithms: linear search, binary search.
- Learning sorting algorithms: bubble sort, quicksort, mergesort.
- Introduction to recursion and its applications.
- Exploring dynamic programming and problem-solving strategies.
- Hands-on: Implementing sorting and searching algorithms in Python, Java, and C++.



Module 5: Database & File Handling

- Introduction to databases, SQL, and relational database concepts.
- Writing SQL queries for data retrieval, insertion, updating, and deletion.
- Working with file handling: reading/writing files in Python, Java, and C++.
- Managing structured and unstructured data in programming.
- Understanding database transactions and normalization.
- Hands-on: Creating a simple CRUD application using Python and SQL.



Module 6: Project Development & Certification

- Learning best practices for structuring and maintaining code.
- Implementing debugging and testing techniques for optimized performance.
- Exploring version control using Git and GitHub.
- Developing a mini-project integrating all learned concepts.
- Code review, performance optimization, and final project assessment.
- Hands-on: Completing a certification exam and receiving final certification.



Training Schedule & Format



The training schedule is structured as a 6-day workshop, providing an intensive and hands-on learning experience. Each day focuses on a key programming module, starting with fundamental concepts and progressing to advanced topics like data structures, algorithms, and database management. The workshop includes interactive sessions, coding exercises, and real-world problem-solving to reinforce learning. Participants will work on mini-projects and hands-on assignments daily, ensuring practical application of concepts. The final day is dedicated to project development, code optimization, and a certification assessment, equipping learners with industry-ready programming skills.

Day	Module	Focus Area	Duration
Day 1	Introduction to Programming	Fundamentals	5hrs
Day 2	Programming Basics	Variables, Loops	5hrs
Day 3	Object-Oriented Programming (OOP)	Classes, Inheritance	5hrs
Day 4	Data Structures & Algorithms	Sorting, Searching	5hrs
Day 5	Database & File Handling	File Handling	5hrs
Day 6	Project Development & Certification	SQL, Project	5hrs

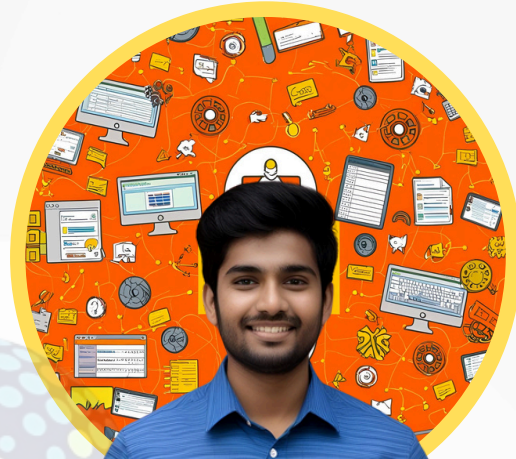
Through hands-on exercises and case studies, participants will develop essential programming skills, explore best practices, and tackle real-world coding challenges. The workshop includes live demonstrations, guided projects, and interactive activities to enhance learning. A collaborative format encourages discussions, problem-solving, and peer learning. By the end, attendees will have a strong programming foundation, enabling them to write efficient code, optimize performance, and build practical applications.

How to Get Involved



Steps to Enroll in Training Programs

- Contact us via email or phone.
- Share your preferred course, schedule, and any specific requirements.
- Our team will review your request and confirm the booking.
- You will receive confirmation details via email or call .



Contact Information

For any questions about the training program, please reach out to:

**Phone**

+91 8869832559

**Website**

<http://philomathnetworks.com>

**Email**

info@philomathnetworks.com

